Applied Linear Algebra

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Outline

System of Linear Equations

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System of Linear Equations

- A **linear equation** in the n variables $x_1, \ldots x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \ldots a_nx_n = b$$

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System of Linear Equations

- A **linear equation** in the n variables $x_1, \ldots x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \ldots a_nx_n = b$$

- A **system** of m linear equations in n variables is a collection of equations of the form:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots a_{1n}x_n & = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots a_{2n}x_n & = b_2 \\ a_{31}x_1 + a_{32}x_2 + \ldots a_{3n}x_n & = b_3 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots a_{mn}x_n & = b_m \end{cases}$$

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- A system of linear equations can be expressed very compactly as $Ax = b$.

- Eg:

$$\begin{cases} -x_1 - x_2 + 2x_3 & = 1 \\ 2x_1 + x_2 + 3x_3 & = 1 \end{cases}$$

$$\begin{bmatrix} -1 & -1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \iff \underbrace{\begin{bmatrix} \dots & R_1^T & \dots \\ \dots & R_2^T & \dots \end{bmatrix}}_{A:coefficientmatrix} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x:variable} = \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{b}.$$

$$\iff x_1 \begin{bmatrix} -1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

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- Depending on the coefficients and b the system:
  - no solution
  - one solution
  - many solutions

- we'll see how to solve linear equations later

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Examples

- The system

$$x_1 + x_2 = 1 \qquad x_1 = -1 \qquad x_1 - x_2 = 0$$

is written as $Ax = b$ with:

has no solution; overdetermined system: tall matrix

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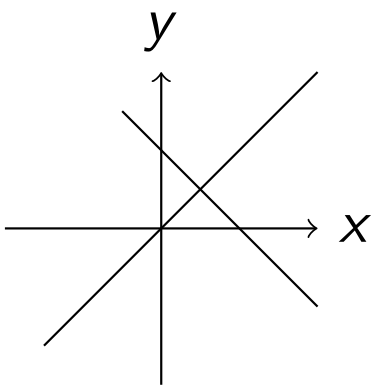
Examples

- The set of linear equations $x_1 + x_2 = 1 \qquad x_2 + x_3 = 2$ is written as $Ax = b$ with

has multiple solutions; undetermined system: wide matrix

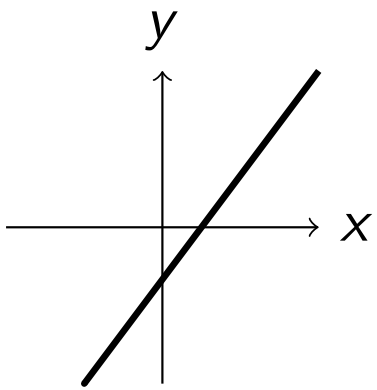
More Examples

$$\begin{cases} x - y &= 0 \\ x + y &= 1 \end{cases}$$



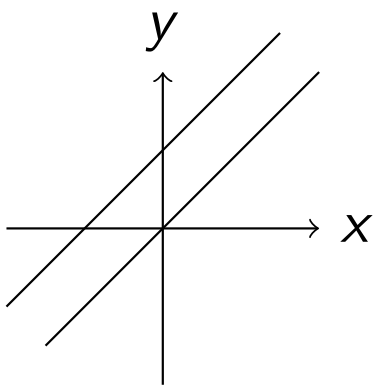
Unique solution

$$\begin{cases} x - y &= -1 \\ 2x - 2y &= -2 \end{cases}$$



Infinitely many solutions

$$\begin{cases} x - y &= -1 \\ x - y &= 0 \end{cases}$$



No solution

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The Column Vectors of A

- $\mathrm{Col}(A) = \{$all linear combinations of the columns of $A\}$
- $Ax = b$ is solvable $\iff$ $b \in \mathrm{Col}(A)$.

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Set of all linear combinations of the column vectors of a matrix A

- The set of all linear combinations of the column vectors of A form a vector space
- What is a vector space?

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Set of all linear combinations of the column vectors of a matrix A

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A vector space always contains the zero vector.

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Find $Col(A)$

- $$A_1 = \begin{bmatrix} 1 & 3 & 8 \\ 1 & 2 & 6 \\ 0 & 1 & 2 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 2 & 5 \\ 1 & 2 & 5 \end{bmatrix}$$

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- Given:

$$\begin{cases} x_1 + 2x_2 + x_3 = 2 \\ 2x_1 + 3x_2 + x_3 = 5 \\ 6x_1 + 4x_2 + x_3 = 8 \end{cases}$$

- $$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 6 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}$$

- How do we solve?

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What if we have an upper triangular matrix?

- $$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}$$

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What if we have an upper triangular matrix?

- $$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix}$$

- We proceed by back substitution

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Goal:

- Transform A into an upper triangular matrix U

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- Transform A into an upper triangular matrix U
- How? By elimination
- Then find the solution by back substitution.
- $$\underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 6 & 4 & 1 \end{bmatrix}}_{\text{A}} \rightarrow \underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{U: after Elimination}}$$

How to get to U:

- One way of doing it is by Gauss Elimination:

- Gauss Elimination:
  - sequence of elementary row operations transforming a matrix A into an upper triangular matrix U
  - We also perform these operations on b, so we put the matrix in what so called an augmented matrix

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- Elementary Row operations consist of:
  - multiplication of a row by a nonzero scalar
  - add a multiple of one row to another
  - change row order.

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Solve our example

$$\underbrace{\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 3 & 1 & 5 \\ 6 & 4 & 1 & 8 \end{array}\right]}_{\text{Augmented Matrix}} \rightarrow \underbrace{\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & -4 \end{array}\right]}_{\text{after Gaussian Elimination}}$$

Algorithm

To solve a system of linear equations:

1. set up the augmented matrix;
2. do Gauss Elimination on the augmented matrix;
3. do backwards substitution.

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Example: Gauss Elimination

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

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